

Problem Set 1: Consumer theory for research

Due date: May 12, 2026

Problem 1

Consider a consumer with preferences that are complete, transitive, monotonic, and convex.

- a) Explain briefly how monotonicity and convexity restrict observable choices.
- b) Provide one concrete example of observed choices that violates: (i) monotonicity; (ii) convexity.
- c) Do violations of these assumptions necessarily imply irrational behavior? Explain briefly.

Problem 2

Consider the utility function:

$$U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}, 0 < \alpha < 1$$

with prices $p_1, p_2 > 0$, income $m > 0$, and utility target $\bar{U} > 0$.

- a) Solve the utility maximization problem and derive the Marshallian demands.
- b) Derive the indirect utility function $v(p_1, p_2, m)$ and show that (i) $v(p_1, p_2, m)$ is increasing in income m , (ii) $v(p_1, p_2, m)$ is decreasing in prices; and (iii) $v(p_1, p_2, m)$ is homogeneous of degree zero in (p_1, p_2, m) .
- c) Solve the expenditure minimization problem and derive the Hicksian demands.
- d) Derive the expenditure function $e(p_1, p_2, \bar{U})$.
- e) Explain briefly how Marshallian demand, Hicksian demand, indirect utility, and expenditure are connected.

Problem 3

Consider a consumer choosing between different subscription plans for a digital platform.

- a) Describe the decision problem: Who is the decision-maker? What choice is being made? What trade-offs does the consumer face?
- b) Propose a simple utility function and briefly explain each component. Your specification should clearly distinguish between observed attributes and unobserved factors.
- c) State one key assumption and explain its role in the model. How would the interpretation of the model change if this assumption does not hold?
- d) Describe one observable implication: What pattern would you expect to observe in actual consumer choices? What kind of differences in the data would allow you to evaluate this prediction?